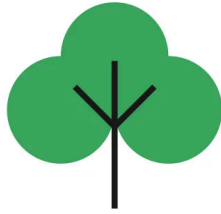




Yagé Botanicals Cloud Migration Proposal



Yagé Botanicals

Network Modernization Proposal

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Step 1: Identify assets



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Action Plan:

1. Establish a Configuration Management Database (CMDB):

Yagé currently lacks formal documentation of its hardware and software. To address this, we recommend deploying a cost-effective CMDB solution (e.g., **iTop**, **Snipe-IT**, or a lightweight asset tracker integrated with Azure or Microsoft 365). The CMDB will log servers, workstations, software licenses, user accounts, and network devices across all five sites. This database will also support future asset lifecycle management and help identify systems that are redundant or non-compliant.

2. Run Automated Asset Discovery Tools:

Use industry-standard tools like **Microsoft Endpoint Manager**, **Nmap**, or **Open-Audit** to perform a full network scan. These tools will detect all active hardware, operating systems, and services, especially important given the presence of outdated and undocumented servers. This step replaces guesswork with verified data and will help prioritize which assets need upgrading, patching, or decommissioning.

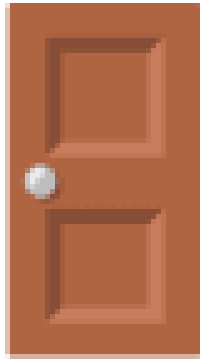
3. Categorize Assets by Criticality and Risk:

After collection, classify assets into categories (e.g., ERP server, domain controllers, personal workstations) and assign risk levels based on exposure and business function. For example:

- Critical assets: Accounting servers, AD Domain Controllers, inventory databases.
- Moderate assets: RODCs at branch sites, network printers, shared drives.
- Low-risk assets: End-of-life machines used only for printing or legacy design tools.

This prioritization ensures time and money are directed where they have the most impact, protecting business continuity and improving appeal to potential buyers.

Step 2: Implement basic security measures



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Action Plan:

1. Apply Critical Security Patches Across Servers and Workstations:

Many of Yagé's systems are running outdated operating systems and applications with known vulnerabilities. Using tools like **Windows Server Update Services (WSUS)** or third-party patch managers, the IT team should immediately:

- Deploy critical and security updates to Windows servers (especially domain controllers and ERP systems).
- Prioritize patching unpatched or unsupported systems, particularly anything running Windows Server 2008 or earlier.
- Maintain a patch log as part of the CMDB to demonstrate compliance readiness.

2. Enforce Strong Authentication for Admin and Critical Accounts:

Basic passwords are insufficient. Yagé should:

- Implement two-factor authentication (2FA)** for privileged accounts and key systems (e.g., domain controllers, file servers).
- Update group policy to enforce **strong password complexity and rotation rules**.
- Immediately disable or audit unused accounts and shared credentials, especially in Active Directory.

3. Deploy Network Segmentation Using VLANs:

Yagé's flat network is vulnerable to lateral movement in case of compromise. As a foundational control:

- Segment the network using VLANs to isolate business-critical servers (e.g., accounting, inventory systems) from user devices and guest traffic.
- Use Access Control Lists (ACLs) on switches and routers to restrict unnecessary communication between VLANs.

4. Implement Basic Endpoint Protection:

Require antivirus and endpoint protection across all endpoints, particularly those used by staff in greenhouses and warehouses. If cost is a factor:

- Begin with **Windows Defender** on Windows 10/11 systems, or deploy lightweight cloud-managed solutions such as **CrowdStrike Falcon Free**, **Sophos Home**, or **Bitdefender Free** for initial coverage.

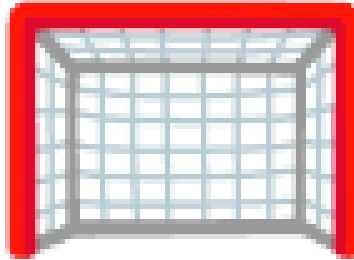
5. **Restore and Reactivate Perimeter Firewall Subscriptions:**

Yagé's home office firewall subscription expired 18 months ago. This must be corrected immediately:

- Renew the **UTM firewall subscription** to restore intrusion prevention (IPS), web filtering, and VPN functionality.
- Update firewall firmware and review rulesets to block unused ports and allow only essential inbound traffic.



Step 3: Strategize the migration



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Action Plan (with Estimated Timelines)

1. Conduct a Cloud Readiness Assessment Using Azure Migrate

Timeline: 1–2 weeks

- Use Azure Migrate to inventory on-premises servers and assess cloud readiness.
- Identify workloads appropriate for rehosting, retirement, or isolation based on technical and business priorities.
- Document system dependencies, application roles, and resource utilization to inform phased migration planning.

2. Prioritize and Migrate Business-Critical Systems

Timeline: 3–5 weeks

- Phase 1 (Weeks 1–2): Migrate the accounting system and file server to Azure Virtual Machines (IaaS) for improved performance and accessibility.
- Phase 2 (Weeks 3–4): Transition ERP components and core business databases to the cloud.
- Phase 3 (Week 5): Decommission outdated or redundant legacy servers identified during the assessment.

3. Deploy Windows Server 2022 Domain Controller with Azure AD Integration

Timeline: 2 weeks

- Install a new Windows Server 2022 domain controller on supported local hardware.
- Integrate with Azure Active Directory using Azure AD Connect to support hybrid identity management.
- Validate replication, and begin decommissioning legacy Windows Server 2008 domain controllers.

4. Establish a Secure Site-to-Site VPN Gateway Between On-Prem and Azure

Timeline: 1 week

- Deploy Azure VPN Gateway to establish encrypted communication between on-premises infrastructure and cloud-hosted systems.
- Configure secure tunnels and routing between Azure Virtual Network and Yagé's primary site.
- Perform end-to-end testing for connectivity, latency, and access control alignment.

5. Upgrade Branch Site Connectivity and Wi-Fi Infrastructure

Timeline: 2–4 weeks (parallel execution)

- Replace existing 5 Mbps sDSL lines at each of the five growing sites with business-class broadband.
- Install business-grade Wi-Fi access points to ensure stable and secure wireless access across all facilities.
- Confirm VLAN segmentation capabilities and apply network isolation policies in accordance with Step 2 (Implement Basic Security Measures).



Step 4: Establish ongoing security monitoring



Step 4: Establish ongoing security monitoring.

Action Plan:

1. Implement a Consolidated SNOC (Security-Network Operations Center) Model

Considering Yagé Botanicals' limited IT personnel and absence of regulatory compliance obligations, the establishment of a dedicated Security Operations Center (SOC) or Network Operations Center (NOC) would be excessive and financially impractical. Instead, the organization should adopt a **Security-Network Operations Center (SNOC)** model. This hybrid approach allows for:

- Unified monitoring of both network performance and security events.
- Centralized visibility into potential system threats, unauthorized access attempts, and uptime issues.
- Efficient use of existing staff to manage both network health and basic security response under a single operational framework.

2. Leverage Existing Microsoft Defender Security Features

To maximize value without investing in new tools, Yagé should utilize the security capabilities already embedded within its Microsoft-based infrastructure:

- Activate **Microsoft Defender Security Center** on all Windows 10/11 endpoints to monitor threat activity and generate alerts.
- Configure **event forwarding** from critical systems (e.g., domain controllers, file servers) to a centralized log repository.
- Maintain a daily log review protocol and define thresholds for alert escalation to internal or external IT support as needed.

3. Integrate Microsoft Sentinel for Cloud-Based Log Analytics (Optional Expansion)

As systems migrate to Azure (as outlined in Step 3), it is recommended to selectively implement **Microsoft Sentinel**, a cloud-native security information and event management (SIEM) solution. Sentinel may be used to:

- Monitor Azure-hosted virtual machines, storage accounts, and Active Directory.
- Correlate logs and generate real-time alerts using built-in analytics rules.
- Control costs by ingesting logs only from high-value systems.

4. **Introduce Selective SOAR Capabilities for High-Risk Scenarios**

While a full Security Orchestration, Automation, and Response (SOAR) platform is not warranted at this time, Yagé Botanicals should implement basic automation using available tools:

- Utilize Microsoft Sentinel's built-in **automated playbooks** (via Logic Apps) to respond to critical events , for example, disabling compromised accounts or alerting IT staff of potential breaches.
- Establish response workflows for common alerts, enabling the IT team to act swiftly without manual intervention.

5. **Develop and Maintain an Incident Response Protocol**

To reinforce the monitoring process and support operational readiness:

- Establish a concise **incident response plan**, including classification of alert severity, containment procedures, and escalation paths.
- Ensure logs and forensic data are retained in accordance with industry best practices (minimum 90-day retention).
- Define roles and responsibilities clearly, including when to involve third-party incident response resources.



Step 5: Stabilize ongoing business operations



Step 5: Stabilize ongoing business operations.

Action Plan:

1. **Formalize IT Policies and Standard Operating Procedures (SOPs):**
 - Develop a written IT operations manual covering system maintenance, user access management, and software update processes.
 - Implement a password policy covering complexity requirements, password rotation, and administrator credential handling.
 - Create a device usage policy to define acceptable use for company-owned and personal (BYOD) devices.

2. **Implement Routine Backup and Recovery Procedures:**
 - Schedule automated, incremental backups for critical systems, including accounting, file storage, and inventory data.
 - Perform quarterly backup restoration tests to ensure data recoverability.
 - Store backups securely offsite or in encrypted cloud storage with access controls.

3. **Introduce Centralized IT Asset Lifecycle Management:**
 - Track hardware and software assets through the CMDB, including purchase dates, warranty status, and assigned users.
 - Maintain software license inventories and set reminders for renewals.
 - Establish procurement and decommissioning standards for all IT equipment.

4. Establish a Basic IT Change Management Process:

- Require logging of all major changes to infrastructure, such as server updates, network changes, or system upgrades.
- Maintain a centralized change log detailing the date, nature, and responsible party for each change.